

Quiz-1

Suppose you are given an integer array. Now write a Java program where you need to modify the same array in such a way that the new elements of each index will be the summation of all the other elements of the array except the current index element and print the largest element.

[You don't need to take the array length and array elements as user input, but your code should work for integer arrays of any length. Also, you are not allowed to create any new array.]

Given Array	Output
<code>int [] arr = {5, 7, 2, 3, 6, 4};</code>	Largest: 7 Updated Array Elements: Index 0: 22 Index 1: 20 Index 2: 25 Index 3: 24 Index 4: 21 Index 5: 23

Suppose you are given an integer array. Now write a Java program where you need to modify the same array in such a way that the new elements of each index will be the product of all the other elements of the array except the current index element. and print the smallest element.

[You don't need to take the array length and array elements as user input, but your code should work for integer arrays of any length. Also, you are not allowed to create any new array.]

Given Array	Output
<code>int [] arr = {5, 2, 3, 6, 4};</code>	Smallest: 2 Updated Array Elements: Index 0: 144 Index 1: 360 Index 2: 240 Index 3: 120 Index 4: 180

Quiz-2

Design the **Student** class to get the desired output as shown below. [You can assume a student will not have more than 5 courses]

```
public class Tester{
    public static void main(String [] args) {
        Student s1 = new Student("Bob");
        Student s2 = new Student();
        Student s3 = new Student("David");
        s2.name = "Carol";
        System.out.println("=====");
        s1.add_course(new String[]{"BUS102", "MKT201",
"MGT211", "MAT101"});
        System.out.println("=====");
        s2.add_course(new String[] {"CSE110", "CSE230",
"CSE260"})
        System.out.println("=====");
        s3.add_course(new String[] {"CSE110", "CSE230",
"CSE260", "MAT110", "PHY111"});
        System.out.println("=====");
        System.out.println(s1.details());
        System.out.println("=====");
        System.out.println(s2.details());
        System.out.println("=====");
        System.out.println(s3.details());
        System.out.println("=====");
        System.out.println(s1.makeGroup(s2));
        System.out.println("=====");
        System.out.println(s1.makeGroup(s1));
        System.out.println("=====");
        System.out.println(s1.makeGroup(s3, s2));
    }
}
```

Output:

```
=====
Bob has 4 courses
Status: Ok
=====
Carol has 3 courses
Status: 1 more course should be added
=====
David has 5 courses
Status: At least 1 course must be dropped
=====
Student name: Bob Program: CSE
Courses: BUS102, MKT201, MGT211,
MAT101
=====
Student name: Carol Program: CSE
Courses: CSE110, CSE230, CSE260
=====
Student name: David Program: CSE
Courses: CSE110, CSE230, CSE260,
MAT110, PHY111
=====
Minimum 3 members needed for the group
=====
You can not make a group with yourself
=====
Bob made a group with Carol and David
```

Design the **Team** class to get the desired output as shown below. [You can assume a team will not have more than 4 members]

```
public class Tester{
    public static void main(String [] args) {
        Team t1 = new Team ("Team A");
        Team t2 = new Team ();
        Team t3 = new Team ("Team C");
        t2.name = "Team B";
        System.out.println("=====");
        t1.add_Member(new String[]{"Alice", "Bob", "Carol"});
        System.out.println("=====");
        t2.add_Member(new String[] {"Dominic", "Emory"})
        System.out.println("=====");
        t3.add_Member(new String[] {"Frankie", "Harper",
        "Isabella", "Josie"});
        System.out.println("=====");
        System.out.println(t1.details());
        System.out.println("=====");
        System.out.println(t2.details());
        System.out.println("=====");
        System.out.println(t3.details());
        System.out.println("=====");
        System.out.println(t1.mergeTeam(t2));
        System.out.println("=====");
        System.out.println(t1.mergeTeam(t1));
        System.out.println("=====");
        System.out.println(t1.mergeTeam(t3, t2));
    }
}
```

Output:

```
=====
Team A has 3 members
Status: Ok
=====
Team B has 2 members
Status: 1 more member should be added
=====
Team C has 4 members
Status: At least 1 member must be dropped
=====
Team name: Team A University: Bracu
Members: Alice, Bob, Carol
=====
Team name: Team B University: Bracu
Members: Dominic, Emory
=====
Team name: Team C University: Bracu
Members: Frankie, Harper, Isabella, Josie
=====
Need at least 3 teams to merge
=====
Can not merge a team with a same team
=====
Team A is now merging with Team B and
Team C
```

Quiz-3

```
public class QuizC{
    public int [] z = {6, 5, 4};
    public int y = -4, sum;
    public int methodA(int x){
        z[2] = z[0] + x;
        y = this.y + z[1];
        sum = sum + 2 + z[0]++;
        x = z[2] + y + x;
        System.out.println(x + " " + y + " " + sum);
        return x;
    }
    public void methodB(int y, int n){
        int x = 0;
        this.y = y + (++n);
        x = x + 2 + n;
        sum = this.sum + methodA(sum) + sum++;
        System.out.println(x + " " + z[0] + " " + sum);
    }
}
```

Illustrate the output of the following statements:

```
QuizC t1 = new QuizC();
```

```
t1.methodB(4, 9);
```

```
t1.methodA(3);
```

=====

```
QuizD t1 = new QuizD();
```

```
t1.methodB(4, 8);
```

```
t1.methodA(5);
```